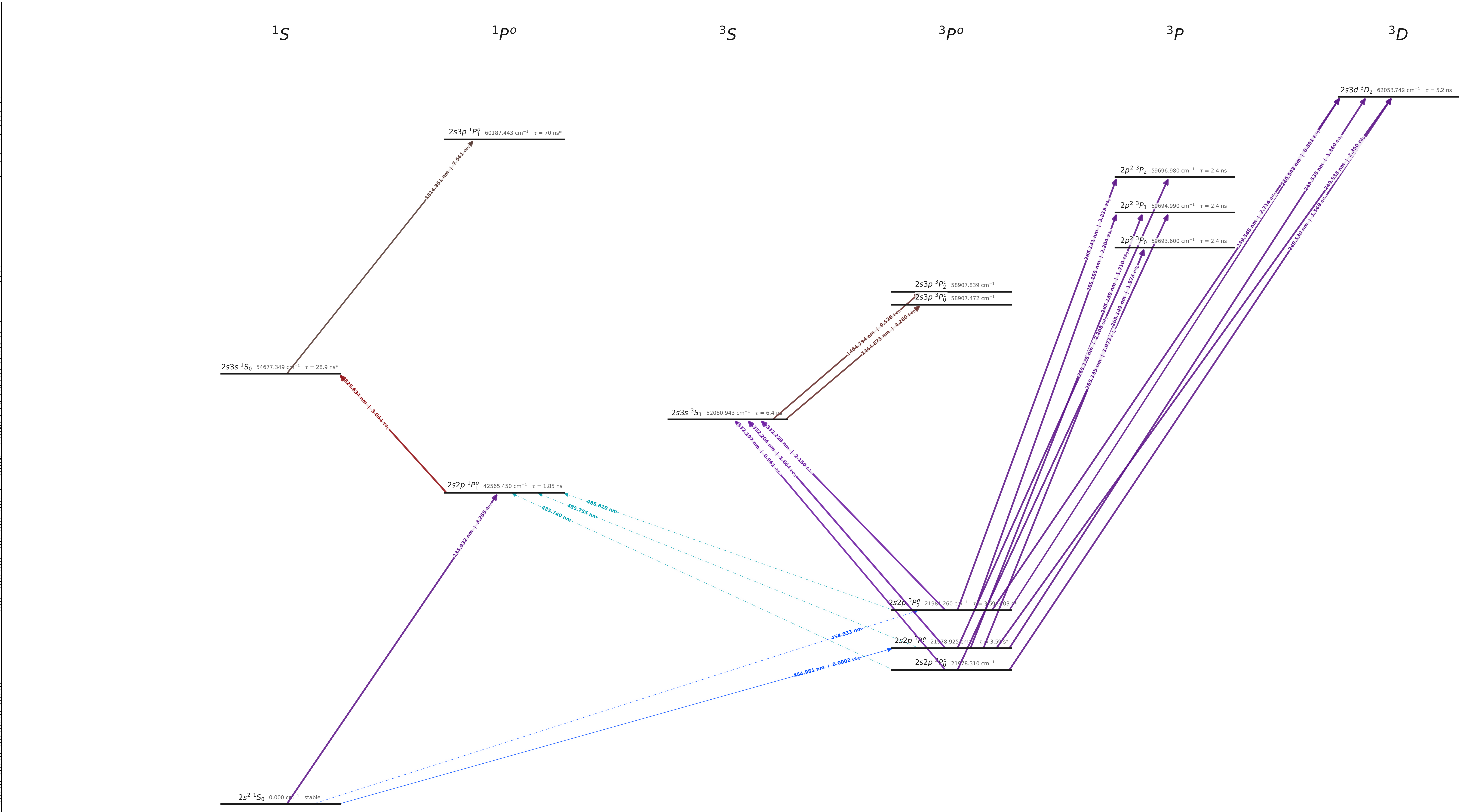


# Beryllium-9

<sup>9</sup>Be (Be I), nuclear spin  $I = 3/2$ . Energy levels and transitions below 2 μm: 15 levels, 24 lines.

Be<sup>+</sup> ionisation limit 75192.64 cm<sup>-1</sup> = 9.32270 eV (132.992 nm)



## Key transitions

| Transition                    | λ vac (nm) | λ air (nm) | ν (THz)       | U  er  J'  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                            |
|-------------------------------|------------|------------|---------------|-------------------------------|----------------------|----------------|------------|--------------------------------|
| $2s^2\ ^1S_0 - 2s2p\ ^1P^o_1$ | 234.9323   | 234.8604   | 1276.08008(5) | 3.255                         | 5.520e+08            | 86.03          | 100        | 234.9 nm resonance line        |
| $2s^2\ ^1S_0 - 2s2p\ ^3P^o_2$ | 454.9813   | 454.8538   | 658.9116      | 0.000                         | 4.200e-01            | 4.43e-08       | 100        | 454.9 nm intercombination line |
| $2s2p\ ^1P^o_1 - 2s3s\ ^1S_0$ | 825.6342   | 825.4073   | 363.10565(13) | 3.064                         | 3.380e+07            | 5.517          | 97.5       | 825.6 nm                       |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>9</sup>Be hyperfine structure ( $I = 3/2$ )

| Level           | A (MHz)       | B (MHz)  | Shift of each F level from the centre of gravity (MHz)     |
|-----------------|---------------|----------|------------------------------------------------------------|
| $2s2p\ ^3P^o_1$ | -139.373(12)  | -0.75(4) | F=1/2: +347.49 F=3/2: +140.13 F=5/2: -209.25               |
| $2s2p\ ^3P^o_2$ | -124.5368(17) | 1.429(8) | F=1/2: +561.67 F=3/2: +373.61 F=5/2: +61.38 F=7/2: -373.25 |
| $2s2p\ ^1P^o_1$ | -13.836       | 0.845    | F=1/2: +35.65 F=3/2: +12.99 F=5/2: -20.54                  |
| $2s3s\ ^3S_1$   | -332.250      | -        | F=1/2: +830.62 F=3/2: +332.25 F=5/2: -498.38               |
| $2p^2\ ^3P_1$   | 18.101        | 0.734    | F=1/2: -44.34 F=3/2: -18.84 F=5/2: +27.34                  |
| $2p^2\ ^3P_2$   | 3.386         | -1.468   | F=1/2: -16.52 F=3/2: -10.16 F=5/2: -0.78 F=7/2: +9.79      |
| $2s3p\ ^1P^o_1$ | -6.704        | 0.427    | F=1/2: +17.29 F=3/2: +6.28 F=5/2: -9.95                    |
| $2s3d\ ^3D_1$   | 150.600       | 0.016    | F=1/2: -376.48 F=3/2: -150.62 F=5/2: +225.90               |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                    | λ vac (nm) | Isotopes | Shift (MHz) |
|-------------------------------|------------|----------|-------------|
| $2s^2\ ^1S_0 - 2s2p\ ^1P^o_1$ | 234.932    | 11-9     | 16015.29(3) |

Shift = ν(first isotope) – ν(second isotope).

Arrow label: vacuum wavelength | reduced dipole matrix element  $|U||er||J'|$  in ea<sub>0</sub>

Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .

Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

